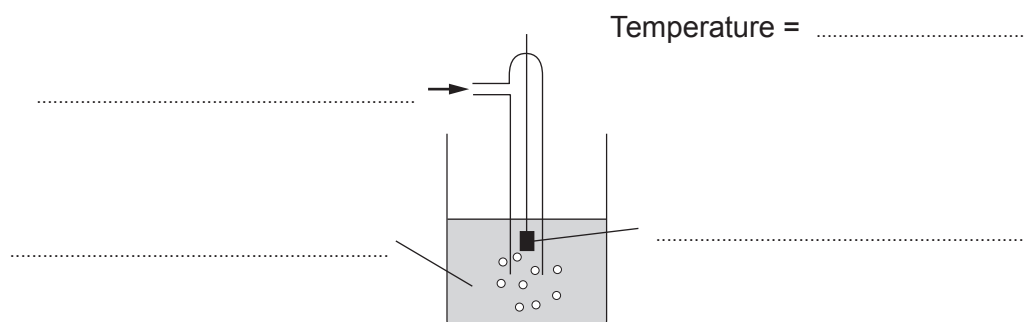


4. Label the diagram of the standard hydrogen electrode below, including all relevant conditions. [2]



5. Reaction rates can be measured using an iodine clock reaction. The data below shows the results obtained by a student in an iodine clock experiment. The concentration of iodide ions was the only difference between the experiments.

Concentration of iodide, $[I^-] / \text{mol dm}^{-3}$	Time taken for colour change / s				Mean time taken / s
	1 st run	2 nd run	3 rd run	4 th run	
0.200	35	46	36	34	
0.400	18	17	18	17	

- (a) Complete the table with appropriate values of mean time taken. [1]
- (b) Calculate the rate of reaction for these two concentrations. [1]

Rate for $0.200 \text{ mol dm}^{-3} [I^-] = \dots\dots\dots \text{s}^{-1}$

Rate for $0.400 \text{ mol dm}^{-3} [I^-] = \dots\dots\dots \text{s}^{-1}$

